

Isabella DeMarti

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Education

Tecnológico de Costa Rica

B.S. in Industrial Production Engineering. GPA: 88.89/100 | 122 credits completed

Campus Local San Carlos

Expected February 2028

Relevant Coursework: Production Planning & Control, Quality Control, Operations Research, Facility Planning & Design, Supply Chain & Logistics, Probability & Statistics I–II, Industrial Computing, Economic Analysis.

Honors: Academic Honor Scholarship for outstanding academic performance (II-2025, I-2026); International Academy for Quality (IAQ) – Best Quality Control Project Award (2026).

Engineering Research Experience

Flexible Manufacturing Cell Project FlexLAB — TEC

San Carlos, Costa Rica

Research and Project Assistant

February 2025 - present

- Restored and operated a FESTO flexible manufacturing cell producing solid and liquid product variants (60–228 units per shift), restructuring effective production availability from 22 to 16.5 hours/week by separating setup, breaks, documentation, and quality activities.
- Built a Microsoft Power Platform OEE pipeline (Microsoft Lists, Power Apps, Power Query) that replaced roughly three weeks of manual Excel consolidation with automated per-shift reporting.
- Developed an automated data architecture integrating production, downtime and quality data into OEE, MTTR and KPI reporting.

Additional Experience

Advanced Manufacturing Laboratory SIMTEC — TEC

San Carlos, Costa Rica

Lab Assistant

February 2024 - present

- Operated and supervised advanced manufacturing equipment (CNC lathe, CNC milling machine, laser cutter, 3D scanners, 3D printers) for course student groups.
- Delivered lab introduction and equipment demonstrations to groups of 20+ visiting students from rural areas.
- Authored a CNC lathe operating manual (specifications, control-panel use, safety guidelines, step-by-step procedures), supporting standardized student use and safer equipment operation.

Leadership & Activities

IV International Meeting of Industrial Engineers

San Carlos, Costa Rica

Organizing Committee

August 2025 - April 2026

- Coordinated a two-day international symposium with 21 speakers (6 international) and 100+ attendees.

International Research Presentation

Bogotá, Colombia

Project Representative / Presenter

December 2025

- Lead-authored and presented research on the enablement of OEE measurement through operational standardization, benchmarking Industry 4.0 laboratories to identify improvements for the FlexLAB project.

Projects and Skills

Award-Winning Quality Improvement Project (DMAIC) – Montik Bakery

June 2026

International Academy for Quality (IAQ) Best Quality Control Project Award (2026)

- Applied DMAIC, statistical analysis, FlexSim simulation and economic evaluation to improve a bakery production process.
- Designed a modular dough-portioning mold projected to reduce portioning time by **40%** while maintaining product quality.

Production Planning & Inventory Control System – Almacigos Santa Clara

June 2026

- Developed an Excel-based production planning system incorporating stage-specific yield-loss modeling, KPI tracking and predictive order-risk alerts.

Facility Layout & Capacity Expansion – Almacigos Santa Clara

June 2026

- Applied Systematic Layout Planning (SLP) to redesign production flow, increasing plant capacity by **20%** without capital investment.

Additional projects: Warehouse Management System (Excel/VBA), Financial Evaluation of FIFCO (FNE, VPN, TMAR), Queueing Theory Analysis (POM-QM).

Technology: Power BI, Power Query (M), Power Apps (Power Fx), Microsoft Lists, Minitab, Excel/VBA, SolidWorks.

Certifications: Certified SOLIDWORKS Associate (CSWA), Dassault Systèmes; Lean Six Sigma Yellow Belt (MoreSteam, in progress); Data Analyst in Power BI (DataCamp, in progress)

Languages: Spanish (Native); English (Bilingual Proficiency / C1)